

Definitions and statement of the main theorem

Let (R, M) be a Noetherian local ring.

Definition 1. R is quasi-complete if for any decreasing sequence $\{A_n\}_{n=1}^\infty$ of ideals of R and each natural number k , there exists a natural number s_k with $A_{s_k} \subseteq (\bigcap_{n=1}^\infty A_n) + M^k$.

Definition 2. R is weakly quasi-complete if for any decreasing sequence $\{A_n\}_{n=1}^\infty$ of ideals of R with $\bigcap_{n=1}^\infty A_n = 0$, and each natural number k , there exists a natural number s_k with $A_{s_k} \subseteq M^k$.

Definition 3. R is analytically irreducible if its M -adic completion $\hat{R}$ is an integral domain.

Quasi-completeness may be viewed as a topological approximation property: any descending chain of ideals eventually “stabilizes modulo M^k ” for each k . The weak variant imposes this only for chains whose intersection is zero. The following basic relationships hold:

Corollary 1. 1. If R is complete (in the M -adic topology), then R is quasi-complete (Chevalley [1]).

2. If R is quasi-complete, then R is weakly quasi-complete.

3. R is quasi-complete if and only if every homomorphic image R/I (for any proper ideal I) is weakly quasi-complete (Anderson [2], Theorem 1.3).

Theorem 1. There exists a weakly quasi-complete Noetherian local ring that is not quasi-complete.

Reduction of the problem

By the third property above, a Noetherian local ring R fails to be quasi-complete if and only if some homomorphic image R/I is not weakly quasi-complete. Therefore, to prove Theorem 1, it suffices to construct a Noetherian local ring A that is weakly quasi-complete but admits a proper quotient that is not weakly quasi-complete.

We now recall two external results that reduce the construction to concrete conditions.

Cited result (Farley). Proposition 1 of Farley, *Quasi-completeness and localizations of polynomial domains* [3] states:

“A Noetherian local integral domain R is weakly quasi-complete if and only if $P \cap R \neq \{0\}$ for each non-zero prime ideal P of $\hat{R}$.”

In other words, R is weakly quasi-complete precisely when the generic formal fiber of R is trivial: the only prime of the completion $\hat{R}$ lying over the zero ideal of R is the zero ideal itself.

Cited result (Anderson). Corollary 2.2 of Anderson, *Quasi-complete semilocal rings and modules* [2] states:

“A one-dimensional Noetherian local domain is (weakly) quasi-complete if and only if it is analytically irreducible.”

Combining these, our task reduces to the following. We seek a Noetherian local domain A such that:

- (A) The generic formal fiber of A is trivial (ensuring A is weakly quasi-complete, by Farley’s criterion).
- (B) There exists a prime element $a \in A$ such that A/aA is a one-dimensional Noetherian local domain that is not analytically irreducible (so A/aA is not weakly quasi-complete, by Anderson’s criterion).

Construction of the complete local domain T

Lemma 1. *Let*

$$T = \mathbb{C}[[x, y, z]]/(x^2 - yz),$$

and let $M = (x, y, z)T$. Then:

1. *T is a complete 2-dimensional Cohen–Macaulay local domain with $|T| = |T/M| = |\mathbb{C}|$.*
2. *The height-one prime ideal $Q = (x, y)T$ is not principal.*

Proof. The ring $\mathbb{C}[[x, y, z]]$ is a complete regular local domain of dimension 3. The quotient

$$T = \mathbb{C}[[x, y, z]]/(x^2 - yz)$$

is isomorphic to the subring $\mathbb{C}[[u^2, uv, v^2]] \subseteq \mathbb{C}[[u, v]]$ via

$$x \mapsto uv, \quad y \mapsto u^2, \quad z \mapsto v^2.$$

Hence T is a domain. Since $x^2 - yz$ is a nonzerodivisor in the regular local ring $\mathbb{C}[[x, y, z]]$, the quotient T is a hypersurface and therefore Cohen–Macaulay of dimension 2.

Also $T/M \cong \mathbb{C}$, and a formal power series ring in finitely many variables over the uncountable field $\mathbb{C}$ has the same cardinality as $\mathbb{C}$, so $|T| = |T/M| = |\mathbb{C}|$.

Finally,

$$T/Q \cong \mathbb{C}[[z]],$$

so Q is prime and has height 1. To see that Q is not principal, note that $MQ \subseteq M^2$, while $x, y \notin M^2$ because the defining relation is quadratic. Hence the images of x and y in Q/MQ are nonzero and linearly independent over $T/M \cong \mathbb{C}$. Thus $\mu(Q) = \dim_{\mathbb{C}}(Q/MQ) \geq 2$, so Q is not principal. $\square$

The two properties of T established above serve different roles in the proof. The cardinality condition $|T| = |T/M|$ is a hypothesis of Jensen’s construction below, which produces the ring A with trivial generic formal fiber (condition (A)). The non-principality of Q will be used to verify condition (B).

Construction of the local UFD A

Lemma 2. *There exists a 2-dimensional local UFD A such that $\hat{A} \cong T$ and the generic formal fiber of A is local with maximal ideal (0) .*

Proof. We apply the following external result.

Cited result (Jensen). Corollary 2.4 of Jensen, *Completions of UFDs with Semi-Local Formal Fibers* [4] states:

“Let (T, M) be a complete local ring and $|T/M| = |T|$. Let $P \in \operatorname{Spec} T$. Then there exists a local UFD A such that $\hat{A} = T$ and the generic formal fiber of A is local with maximal ideal P iff T is a field or DVR and $P = (0)$, or T has depth at least two and:

1. P is nonmaximal and contains all associated prime ideals of T ;
2. $P \cap$ the prime subring of T is (0) ;
3. if $J \in \operatorname{Spec} T$ with $\operatorname{ht}(J) > \operatorname{depth}(T_J) = 1$, then $J \subseteq P$.”

We apply this with the ring T from Lemma 1 and with $P = (0)$. The hypotheses hold:

1. T is complete and $|T| = |T/M|$ by Lemma 1.
2. T has depth 2 by Lemma 1.
3. Since T is a domain, its only associated prime is (0) , so $P = (0)$ contains all associated primes.
4. $P = (0)$ is nonmaximal and meets the prime subring trivially.
5. Because T is Cohen–Macaulay, there is no prime J with $\text{ht}(J) > \text{depth}(T_J) = 1$.

Therefore Jensen’s corollary yields a local UFD A with $\widehat{A} \cong T$ whose generic formal fiber is local with maximal ideal (0) . $\square$

Verification that A is the desired counterexample

By Lemma 2, the generic formal fiber of A is trivial: the only prime ideal of $\widehat{A} \cong T$ contracting to (0) in A is (0) itself. Hence every nonzero prime ideal of $\widehat{A}$ has nonzero contraction to A . By Farley’s criterion, A is weakly quasi-complete. This establishes condition (A).

It remains to verify condition (B). Consider the height-one prime $Q = (x, y)T$ from Lemma 1. Since $Q \neq (0)$, the triviality of the generic formal fiber implies

$$q := Q \cap A \neq (0).$$

Because $\widehat{A}$ is faithfully flat over A , we have

$$1 \leq \text{ht}(q) \leq \text{ht}(Q) = 1,$$

so $\text{ht}(q) = 1$. As A is a UFD, $q = aA$ for some prime element $a \in A$.

We claim that aT is not prime. Indeed, $aT \subseteq Q$. If aT were prime, then $\text{ht}(aT) = 1$, so the inclusion $aT \subseteq Q$ of height-one prime ideals would force $aT = Q$, contradicting the fact that Q is not principal.

Therefore T/aT is not a domain. Since completion commutes with quotient for Noetherian local rings,

$$\widehat{A/aA} \cong \widehat{A}/a\widehat{A} \cong T/aT,$$

so $\widehat{A/aA}$ is not a domain. Because a is prime in the domain A , the quotient A/aA is a one-dimensional Noetherian local domain. Hence A/aA is not analytically irreducible. By Anderson’s criterion, A/aA is not weakly quasi-complete. This establishes condition (B).

Conclusion

The ring A is weakly quasi-complete (condition (A)), but its homomorphic image A/aA is not weakly quasi-complete (condition (B)). Since quasi-completeness is equivalent to every homomorphic image being weakly quasi-complete, A is not quasi-complete. Thus A is a weakly quasi-complete Noetherian local ring that is not quasi-complete, proving Theorem 1. $\square$

References

- [1] Claude Chevalley. On the theory of local rings. *Annals of Mathematics*, 44(4):690–708, 1943.
- [2] Daniel D Anderson. Quasi-complete semilocal rings and modules. In *Commutative Algebra: Recent Advances in Commutative Rings, Integer-Valued Polynomials, and Polynomial Functions*, pages 25–37. Springer, 2014.
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- [4] David Jensen. Completions of ufd’s with semi-local formal fibers. *Communications in Algebra*®^(R), 34(1):347–360, 2006.